Acknowledgements

This work was supported by U.S. Air Force Office of Scientific Research, grants FA8655-05-1-3010, FA9550-15-1-0326, FA9550-18-1-0521, FA9550-22-1-0289, and by the German Federal Ministry of Economics and Technology, grant ZIM KF2341201FR9.

We thank Dr. Thomas Mattsson (Sandia National Laboratories, U.S.A.) for providing QMD and SESAME high-density EOS data of water, and we thank Jean-Paul Jay-Gerin and Jintana Meesungnoen (Université de Sherbrooke, Québec, Canada) for providing data of the attenuation length of electron beams.

Author declarations

Conflict of Interest

The authors have no conflicts to disclose.

Author contributions

Norbert Linz: Conceptualization (supporting), methodology (equal), experiments (equal), data interpretation (equal), writing (supporting), **Sebastian Freidank:** Methodology (equal), experiments (equal), data interpretation (supporting), **Xiao-Xuan Liang:** Methodology (equal), data interpretation (equal), **Alfred Vogel:** Conceptualization (lead), fund raising, methodology (equal), data interpretation (equal), writing (lead).

Data availability

Most data supporting the findings of this study are available within the article. The raw data used for determining the dependence of energy partitioning of absorbed laser energy as a function of average plasma energy density (Fig. 11) are available on request from the authors.

ORCID IDs:

Norbert Linz	https://orcid.org/0000-0002-7843-5966
Sebastian Freidank	https://orcid.org/0000-0002-5453-269X
Xiao-Xuan Liang	https://orcid.org/0000-0002-8325-1627
Alfred Vogel	https://orcid.org/0000-0002-4371-9037

Accepted Manuscript

References

1. Gattass R R and Mazur E 2008 Femtosecond laser micromachining in transparent materials *Nature Photon.* **2** 219-225
2. Minoshima K, Kowalevicz A M, Hartl I, Ippen E P, and Fujimoto J G 2001 Photonic device fabrication in glass by use of nonlinear materials processing with a femtosecond laser oscillator *Opt. Lett.* **26** 1516-1518
3. Schaffer C B, Brodeur A, García J F, and Mazur E 2001 Micromachining bulk glass by use of femtosecond laser pulses with nanojoule energy *Opt. Lett.* **26** 93-95
4. Della Valle G, Osellame R, and Laporta P 2009 Micromachining of photonic devices by femtosecond laser pulses *J. Opt. A* **11** 013001
5. Sugioka K and Cheng Y 2014 Ultrafast lasers—reliable tools for advanced materials processing *Light Sci. Appl.* **3** e149
6. Balling P and Schou J 2013 Femtosecond-laser ablation dynamics of dielectrics: basics and applications for thin films *Rep. Prog. Phys.* **76** 036502
7. Shugaev M V, et al. 2016 Fundamentals of ultrafast laser-material interaction *Mrs Bull* **41** 960-968
8. Stoian R and Colombier J-P 2020 Advances in ultrafast laser structuring of materials at the nanoscale *Nanophotonics* **9** 4665-4688
9. Raciukaitis G 2021 Ultra-Short Pulse Lasers for Microfabrication: A Review *IEEE J. Sel. Topics Quantum Electron.* **27** 1-12
10. Stoian R and Bonse (eds.) J 2023 *Ultrafast Laser Nanostructuring* Springer Series in Optical Sciences
11. Vogel A, Hentschel W, Holzfuss J, and Lauterborn W 1986 Cavitation bubble dynamics and acoustic transient generation in ocular surgery with pulsed neodymium: YAG lasers *Ophthalmology* **93** 1259-1269
12. Vogel A, Schweiger P, Frieser A, Asiyo M, and Birngruber R 1990 Intraocular Nd:YAG laser surgery: damage mechanism, damage range and reduction of collateral effects. *IEEE J. Quantum Electron.* **26** 2240-2260
13. Juhasz T, Loesel F H, Kurtz R M, Horvath C, Bille J F, and Mourou G 1999 Corneal refractive surgery with femtosecond lasers *IEEE J. Sel. Topics Quantum Electron.* **5** 902-910
14. Palanker D V, et al. 2010 Femtosecond laser-assisted cataract surgery with integrated optical coherence tomography *Sci. Transl. Med.* **2** 58ra85
15. Friedman N J, et al. 2011 Femtosecond laser capsulotomy *J. Cataract. Refr. Surg.* **37** 1189-1198
16. Trost A, et al. 2013 A new nanosecond UV laser at 355 nm: Early results of corneal flap cutting in a rabbit model *Invest. Ophthalmol. Vis. Sci.* **54** 7854-7864
17. Linz N, Freidank S, Liang X-X, and Vogel A 2023 *Laser Micro- and Nanostructuring for Refractive Eye Surgery* in Ultrafast Laser Nanostructuring edited by Stoian R and Bonse J (Springer)
18. Koenig K, Riemann I, Fischer P, and Halbhuber K H 1999 Intracellular nanosurgery with near infrared femtosecond laser pulses *Cell. Mol. Biol.* **45** 195-201
19. Venugopalan V, Guerra A, Nahen K, and Vogel A 2002 Role of laser-induced plasma formation in pulsed cellular microsurgery and micromanipulation *Phys. Rev. Lett.* **88** 078103
20. Tirlapur U K and Koenig K 2002 Targeted transfection by femtosecond laser *Nature* **418** 290-291
21. Yanik M F, Cinar H, Cinar H N, Chisholm A D, Jin Y, and Ben-Yakar A 2004 Neurosurgery: Functional regeneration after laser axotomy *Nature* **432** 822-822
22. Vogel A, Noack J, Huettman G, and Paltauf G 2005 Mechanisms of femtosecond laser nanosurgery of cells

Accepted Manuscript

and tissues *Appl. Phys. B.* **81** 1015-1047

23. Rau K R, Quinto-Su P A, Hellman A N, and Venugopalan V 2006 Pulsed Laser Microbeam-Induced Cell Lysis: Time-Resolved Imaging and Analysis of Hydrodynamic Effects *Biophys. J.* **91** 317-329
24. Chung S H and Mazur E 2009 Surgical applications of femtosecond lasers *J. Biophotonics* **2** 557-572
25. Stevenson D J, Gunn-Moore F J, Campbell P, and Dholakia K 2010 Single cell optical transfection *J. R. Soc. Interface* **7** 863-871
26. Ronchi P, Terjung S, and Pepperkok R 2012 At the cutting edge: Applications and perspectives of laser nanosurgery in cell biology *Biol. Chem.* **393** 235-248
27. Hoy C L, Ferhanoglu O, Yildirim M, Kim K H, Karajanagi S S, Chan K M C, Kobler J B, Zeitels S M, and Ben-Yakar A 2014 Clinical ultrafast laser surgery: Recent advances and future directions *IEEE J. Sel. Topics Quantum Electron.* **20** 242-255
28. Ohl C-D, Arora M, Ikink R, de Jong N, Versluis M, Delius M, and Lohse D 2006 Sonoporation from Jetting Cavitation Bubbles *Biophys. J.* **91** 4285-4295
29. Ohl C-D, Arora M, Dijkink R, Janve V, and Lohse D 2006 Surface cleaning from laser-induced cavitation bubbles *Appl. Phys. Lett.* **89** 074102
30. Hellman A N, Rau K R, Yoon H H, Bae S, Palmer J F, Phillips K S, Allbritton N L, and Venugopalan V 2007 Laser-Induced Mixing in Microfluidic Channels *Anal Chem* **79** 4484-4492
31. Zwaan E, Le Gac S, Tsuji K, and Ohl C-D 2007 Controlled Cavitation in Microfluidic Systems *Phys. Rev. Lett.* **98** 254501
32. Guo S X, Bourgeois F, Chokshi T, Durr N J, Hilliard M A, Chronis N, and Ben-Yakar A 2008 Femtosecond laser nanoaxotomy lab-on-a-chip for in vivo nerve regeneration studies *Nat. Methods* **5** 531-533
33. Dijkink R and Ohl C-D 2008 Laser-induced cavitation based micropump *Lab on a Chip* **8** 1676–1681
34. Compton J L, Luo J C, Ma H, Botvinick E, and Venugopalan V 2014 High-throughput optical screening of cellular mechanotransduction *Nat. Photonics* **8** 710-715
35. Reese H, Schädel R, Reuter F, and Ohl C-D 2022 Microscopic pumping of viscous liquids with single cavitation bubbles *J. Fluid. Mech.* **944** A17
36. Linz N, Freidank S, Liang X X, Vogelmann H, Trickl T, and Vogel A 2015 Wavelength dependence of nanosecond infrared laser-induced breakdown in water: Evidence for multiphoton initiation via an intermediate state *Phys. Rev. B* **91** 134114
37. Linz N, Freidank S, Liang X X, and Vogel A 2016 Wavelength dependence of femtosecond laser-induced breakdown in water and implications for laser surgery *Phys. Rev. B* **94** 024113
38. Garcia-Lechuga M, Utéza O, Sanner N, and Grojo D 2023 Wavelength-Independent Performance of Femtosecond Laser Dielectric Ablation Spanning Over Three Octaves *Phys. Rev. Appl.* **19** 044047
39. Bell C E and Landt J A 1967 Laser-induced high-pressure shock waves in water *Appl. Phys. Lett.* **10** 46-48
40. Vogel A, Busch S, and Parlitz U 1996 Shock wave emission and cavitation bubble generation by picosecond and nanosecond optical breakdown in water *J. Acoust. Soc. Am.* **100** 148-165
41. Lauterborn W and Vogel A 2013 *Shock Wave Emission by Laser Generated Bubbles* in Bubble Dynamics and Shock Waves edited by Delale C F (Springer-Verlag Berlin Heidelberg)
42. Liang X X, Linz N, Freidank S, Paltauf G, and Vogel A 2022 Comprehensive analysis of spherical bubble oscillations and shock wave emission in laser-induced cavitation *J. Fluid. Mech.* **940** A5
43. Denner F and Schenke S 2023 Modeling acoustic emissions and shock formation of cavitation bubbles

Accepted Manuscript